AquaPearl 500 Pro

Smart reverse-osmosis water purifier with UV-LED sterilisation

User Manual, Revision C (March 2026). This revision describes purifiers running firmware version 2.2.

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Please read this manual completely before you use the purifier, and keep it for future reference. Installation is described in the separate AquaPearl Installation Guide.

1. About this manual

This manual explains how to operate, maintain and clean the AquaPearl 500 Pro. It covers the control panel, the indicator lights, the Pearlz Home app, filter replacement, cleaning and descaling, and the audible alert codes the purifier uses to tell you that something needs attention.

Installing the purifier, including the first-run flushing procedure, is covered in the AquaPearl Installation Guide. Warranty terms are set out in the Pearlz Warranty Policy, and maintenance plans in the Service & Maintenance Agreement. Where this manual summarises those documents, the full document takes precedence.

1.1 Firmware version

The purifier's behaviour is controlled by firmware that Pearlz updates from time to time over Wi-Fi. This revision of the manual describes firmware version 2.2. Later firmware versions can change how some features behave. You can see the version your purifier is running in the Pearlz Home app under Settings, then Device info.

1.2 Symbols used in this manual

- **WARNING** marks a risk of injury, electric shock or water damage.
- **CAUTION** marks a risk of damaging the purifier.
- **NOTE** marks useful information that is not a safety matter.

2. Safety warnings

WARNING: Connect the purifier to a cold drinking-water supply only. Never connect it to a hot-water line or to water warmer than 38 °C.

- Use the purifier only with water that is already safe to drink, such as municipal mains water. It is not designed to make microbiologically unsafe water, such as untreated river or pond water, safe to drink.
- Plug the purifier only into a grounded three-pin socket rated 220-240 V, 50 Hz. Do not use extension leads or multi-plug adaptors.
- Unplug the purifier and close the feed-water valve before you open the filter compartment, replace a cartridge or clean the unit.
- Do not open the main housing. There are no user-serviceable parts inside it. Opening the housing or having it repaired by anyone other than a Pearlz-authorised technician voids the warranty.

- Do not install or use the purifier outdoors, in direct sunlight, or within 50 cm of a stove, oven or other heat source.
- Do not place heavy objects on top of the purifier or lean on it once it is wall-mounted.
- Children should not play with the purifier. Cleaning and maintenance should not be done by children without supervision.
- If you notice water inside the housing, a burning smell, smoke or sparks, unplug the purifier immediately, close the feed-water valve and contact Pearlz support. Do not use it again until it has been inspected.
- If the power cord is damaged, it must be replaced by Pearlz or an authorised technician.
- Use only genuine Pearlz cartridges and spare parts.

CAUTION: If the purifier has been transported or stored below 0 °C, let it stand at room temperature for at least 4 hours before switching it on. Water frozen inside the membrane can damage it.

3. What's in the box

Check that the box contains the following items before you begin installation:

1. AquaPearl 500 Pro purifier unit, with the PF-1, CB-2, RO-3 and MN-4 cartridges already fitted
2. Wall-mounting bracket with four screws and four wall plugs
3. Feed-water valve (1/2-inch to 1/4-inch ball-valve tee)
4. Inlet tube, 3 metres, white, 1/4-inch
5. Drain tube, 2 metres, red, 1/4-inch, with drain saddle clamp
6. Detachable power cord, 1.5 metres, three-pin plug
7. Quick start card
8. This user manual and the warranty card

The following items are not included and are sold separately: digital TDS meter pen, pressure-reducing valve, spare wall-mount bracket kit, sanitising tablets and replacement cartridges. A booster pump, which is needed where mains pressure is below 1.0 bar, is supplied only as part of a professional installation.

If anything is missing or damaged, contact Pearlz support within 48 hours of delivery and keep all packaging.

4. Technical specifications

Specification	Value
Purification stages	5: PF-1 sediment, CB-2 carbon block, RO-3 reverse-osmosis membrane, MN-4 mineraliser, UV-LED sterilisation
Dispensing flow rate	1.8 litres per minute
Production rate	12 litres per hour
Storage tank capacity	8 litres
Purified to waste water ratio	1 : 1
Dimensions (W x D x H)	260 x 420 x 480 mm
Weight (empty)	11.5 kg
Power supply	220-240 V AC, 50 Hz
Rated power	48 W; standby below 1.5 W

Specification	Value
Inlet water pressure	1.0 to 4.0 bar
Inlet water temperature	5 to 38 °C
Maximum inlet TDS	2,000 ppm
Noise while producing water	45 dB(A) or less
Connectivity	Wi-Fi 2.4 GHz (802.11 b/g/n); Pearlz Home app for Android 9 or later and iOS 15 or later
Display	Purified-water TDS in ppm, filter life, alert codes

4.1 Filtration performance

In Pearlz laboratory tests using feed water at 500 ppm TDS and 25 °C, the AquaPearl 500 Pro reduced total dissolved solids by 92 to 96 percent. The RO-3 membrane has a nominal pore size of 0.0001 micron. The MN-4 mineraliser then adds back small amounts of calcium and magnesium to improve taste, so the purified-water TDS shown on the display is typically between 40 and 90 ppm.

Actual performance depends on the quality, temperature and pressure of your feed water and on the age of the cartridges. Performance figures in this manual are from Pearlz internal testing.

5. Getting to know your purifier

5.1 Front

- **Tank lid**, on top of the unit. Opened only for six-monthly tank sanitisation.
- **Display**, showing the TDS of the purified water, filter life and alert codes.
- **Control panel**, with five touch buttons below the display.
- **Dispensing nozzle**, with a removable nozzle cap.
- **Drip tray**, removable, with a red float that rises when the tray is full.

5.2 Back and sides

- **Inlet port** (white tube) and **drain port** (red tube) on the back panel.
- **Power socket** for the detachable cord.
- **Rating label**, showing the model, serial number and manufacturing date. You will need the serial number for warranty claims and service visits.
- **Filter compartment door** on the right side, held by two magnetic catches.

5.3 Inside the filter compartment

The four cartridges sit side by side in numbered positions, from left to right: PF-1, CB-2, RO-3 and MN-4. Each cartridge clicks into a quick-connect head and is released with a quarter turn. The UV-LED module is built into the tank outlet. It is not visible from the filter compartment and is not a user-replaceable part.

6. Control panel

6.1 Buttons

Button	Tap	Press and hold
POWER	Wake the display	3 seconds: switch the purifier on or to standby
DISPENSE	Dispense 250 ml	Dispense continuously until released
FLUSH	No effect	3 seconds: start a 10-minute flush cycle to the drain
RESET	Show remaining filter life for each cartridge	5 seconds: reset the filter alert currently shown
WI-FI	Show Wi-Fi status	3 seconds: enter pairing mode

6.2 Display

In normal use, the display shows the TDS of the purified water in ppm. Tap RESET to cycle through the remaining life of each cartridge as a percentage. When an alert is active, the display shows its code, for example E-03, and the ALERT light turns red.

6.3 Child lock

Press DISPENSE and RESET together for 3 seconds to lock or unlock the buttons. While the lock is on, the display shows a padlock symbol and DISPENSE does nothing.

7. Indicator lights

Light	State	Meaning
POWER	White, steady	The purifier is on
POWER	Off	The purifier is in standby or unplugged
FILTER	Green	All cartridges are within their service life
FILTER	Amber, steady	A cartridge is due for replacement within 15 days
FILTER	Red, flashing	A cartridge has reached the end of its life, or alert E-03 is active
UV	Blue, steady	UV-LED sterilisation is active
UV	Blue, flashing	The UV-LED module needs attention; contact Pearlz support
WI-FI	Blue, blinking	Pairing mode
WI-FI	Blue, steady	Connected to the Pearlz Home app
WI-FI	Off	Wi-Fi is not set up or is switched off
TANK	Green, steady	The storage tank is full
TANK	Green, blinking	The tank is filling
ALERT	Red, steady	An alert code is shown on the display; see Section 13

NOTE: A FILTER indicator pulsing red with no audible alert means a cartridge's life counter has reached zero. A FILTER indicator pulsing red together with a triple short tone is alert E-03, described in Section 13.

8. Operating your purifier

8.1 Switching on for the first time

Switch on only after the purifier has been installed and the first-run flushing procedure in the Installation Guide has been completed. Hold POWER for 3 seconds. The POWER light turns white, the TANK light blinks green while the tank fills, and the display shows a TDS reading after about two minutes. A full tank takes about 40 minutes to produce.

8.2 Dispensing water

Place a glass under the nozzle and tap DISPENSE for 250 ml, or hold DISPENSE for as long as you need. Water is dispensed at 1.8 litres per minute. If the tank is empty, the purifier dispenses at the production rate, which is much slower.

8.3 How the purifier produces water

The purifier refills its tank automatically whenever the level drops. For every litre of purified water it sends about one litre of concentrate to the drain; this is normal for reverse-osmosis purification. The pump stops as soon as the tank is full. A faint humming sound and occasional clicks from the valves while the tank fills are normal.

8.4 When you go away

If you will be away for more than 14 days, close the feed-water valve, dispense all the water in the tank, and switch the purifier to standby or unplug it. When you return, open the valve, switch on, and run the first-run flushing procedure described in the Installation Guide before drinking the water.

8.5 After a power cut

The purifier restarts automatically when power returns and keeps the water already in the tank. If the power has been off for more than 48 hours, dispense and discard the water in the tank before using it.

9. The Pearlz Home app

9.1 What the app does

The free Pearlz Home app lets you see the remaining life of each cartridge, the TDS history of your purified water and any active alerts. You can also order cartridges, book a service visit and receive firmware updates through it. The purifier works fully without the app; the app is optional.

9.2 Pairing the purifier with the app

1. Install the Pearlz Home app and create an account.
2. Make sure your phone is connected to a 2.4 GHz Wi-Fi network. The purifier cannot connect to 5 GHz-only networks.
3. On the purifier, hold WI-FI for 3 seconds until the WI-FI light blinks blue.
4. In the app, tap Add device, choose AquaPearl 500 Pro and follow the instructions. Keep your phone within 5 metres of the purifier.
5. When pairing succeeds, the WI-FI light turns steady blue. Pairing mode ends automatically after 3 minutes if no phone connects.

9.3 Firmware updates

When the purifier is connected, firmware updates are downloaded and installed automatically overnight. The purifier keeps working normally during an update and restarts once, which takes about one minute. You can see the installed version in the app under Settings, then Device info.

9.4 Your data

The app collects water-usage and device data to calculate filter life and show your history. What is collected, how long it is kept and how to request or delete it is explained in the Pearlz Privacy Policy.

10. Filter system and replacement intervals

10.1 What each stage does

Stage	Cartridge	What it does	Replace every
1	PF-1 sediment pre-filter	Traps sand, rust and particles down to 5 microns; protects the later stages	6 months or 3,000 litres
2	CB-2 carbon block	Reduces chlorine, odours and organic compounds	12 months or 6,000 litres

Stage	Cartridge	What it does	Replace every
3	RO-3 revers e-osmosis membrane	Removes dissolved salts and most dissolved solids	24 months or 12,000 litres
4	MN-4 mineraliser	Adds calcium and magnesium for taste	12 months or 6,000 litres
5	UV-LED module	Sterilises water as it leaves the tank	Not user-replaceable; rated for 10,000 hours

The interval is whichever comes first: the number of months or the number of litres.

10.2 How the purifier tracks filter life

The purifier counts both the litres it has purified and the days since each cartridge was reset, and uses whichever runs out first. Fifteen days before a cartridge is due, the FILTER light turns amber. When a cartridge reaches the end of its life, the FILTER light flashes red. The purifier keeps producing water, but the water quality can no longer be assured.

Very hard or sandy feed water can block the PF-1 pre-filter before its counter runs out. In that case the purifier raises alert E-03 (Section 13) when it detects that flow through the pre-filter has dropped.

10.3 Use genuine cartridges

Use only genuine Pearlz cartridges. Cartridges from other makers may not seal correctly in the quick-connect heads, can leak, and are not covered by the warranty.

11. Filter maintenance

This section explains how to replace cartridges yourself. A Pearlz technician can also replace them during a service visit.

11.1 Before you start

You will need the new cartridge, a towel and a small bowl. No tools are required.

1. Switch the purifier to standby and unplug it.
2. Close the feed-water valve.
3. Hold DISPENSE until water stops, to release the pressure in the system.
4. Place the towel and bowl under the filter compartment.

11.2 Replacing the PF-1 sediment pre-filter

1. Open the filter compartment door on the right side.
2. Hold the PF-1 cartridge in the first position and turn it a quarter turn anticlockwise.
3. Pull it straight down out of its head. A little water will drip out; this is normal.
4. Remove the protective caps from the new PF-1 cartridge.
5. Push the new cartridge straight up into the head and turn it a quarter turn clockwise until it clicks.
6. Close the door, open the feed-water valve and plug the purifier in.
7. Check the cartridge head for drips for two minutes.

11.3 Replacing the CB-2 and MN-4 cartridges

Follow the same steps as for PF-1, using the second position for CB-2 and the fourth position for MN-4. The cartridges are shaped differently and only fit their own positions.

11.4 Replacing the RO-3 membrane

The RO-3 membrane is heavier and seals under higher pressure than the other cartridges. Pearlz recommends that a technician replaces it, but you can do it yourself by following the PF-1 steps in the third position. After a new membrane is fitted, the TDS reading can stay higher than usual for the first two tanks of water.

11.5 Resetting the filter alert

After fitting a new cartridge, hold RESET for 5 seconds. The purifier sounds a single confirmation tone and resets the counter for the cartridge whose alert is currently shown. If you replaced a cartridge that is not currently showing an alert, or more than one cartridge, reset each one from the app under Filters so that the other counters are kept. If you do not use the app, a technician can reset individual counters during a service visit.

11.6 Flush before drinking

Every new cartridge must be flushed before you drink the water. Run the first-run flushing procedure described in Section 6 of the AquaPearl Installation Guide after every cartridge replacement.

12. Cleaning and descaling

Regular cleaning keeps the purifier hygienic and the water tasting fresh. Always unplug the purifier before cleaning it.

Task	How often
Wipe the exterior	Every week
Empty and rinse the drip tray	When the red float rises, and at least every week
Descale the dispensing nozzle	Every 3 months
Sanitise the storage tank	Every 6 months, ideally when you replace the PF-1 pre-filter

12.1 Exterior

Wipe the housing and display with a soft cloth dampened with clean water. Do not use abrasive pads, bleach, alcohol or solvent-based cleaners, which can damage the display and the plastic.

12.2 Drip tray

Slide the drip tray out, empty it, rinse it under running water and dry it before refitting. If the tray fills more than once a day, the nozzle may be dripping; descale it as described below.

12.3 Descaling the dispensing nozzle

Mineral scale can build up on the nozzle and make it drip.

1. Unscrew the nozzle cap by turning it anticlockwise.
2. Soak it for 20 minutes in a solution of equal parts white vinegar and water.
3. Brush it gently with a soft toothbrush and rinse it thoroughly in drinking water.
4. Screw the cap back on and dispense and discard one glass of water.

12.4 Sanitising the storage tank

1. Dispense all the water in the tank, then unplug the purifier and close the feed-water valve.
2. Dissolve one Pearlz sanitising tablet in 1 litre of purified water.
3. Open the tank lid and pour the solution into the tank. Close the lid.
4. Wait 30 minutes.

5 Plug in, open the feed-water valve and dispense all the water in the tank to a sink, then discard it.

6 Before drinking the water again, run the first-run flushing procedure in Section 6 of the AquaPearl Installation Guide.

CAUTION: Never use household bleach, detergent or vinegar inside the tank. Use only Pearlz sanitising tablets.

13. Audible alert codes

When the purifier detects a problem, it sounds an audible alert pattern, turns the ALERT indicator red and shows a code on the display. The table lists the codes used by firmware version 2.2.

Code	Audible alert pattern	Meaning	What to do
E-01	Single long tone	Water supply interrupted: no inlet water for 60 seconds	Check that the feed-water valve is open and the inlet tube is not kinked. The purifier resumes automatically when water returns.
E-02	Double short tone, repeated every minute	Leak detected: the leak sensor in the base is wet. The purifier closes its inlet valve.	Unplug, close the feed-water valve and dry the base. If there is water inside the housing, or the code returns after drying, contact Pearlz support. Do not open the housing.
E-03	Triple short tone; FILTER indicator pulses red	Sediment pre-filter (PF-1) saturation: flow through the pre-filter has dropped	Replace the PF-1 cartridge (Section 11.2), then hold RESET for 5 seconds to clear the alert.
E-04	Quadruple short tone	RO membrane output low	Check the TDS reading. If it stays above 100 ppm, or the tank takes more than 90 minutes to fill, replace the RO-3 membrane.
E-05	Continuous tone for 10 seconds	Pump overheated: the pump ran for more than 30 minutes without filling the tank	Unplug for 30 minutes and check the inlet water pressure. If E-05 appears again, contact Pearlz support.
E-06	Double long tone	Inlet water pressure too low: below 1.0 bar	Check whether other taps have low pressure. If your mains pressure is regularly below 1.0 bar, a booster pump is needed; book a professional installation.
E-07	Rapid five-pulse tone	Tank level sensor fault	Switch the purifier off and on again. If the code returns, contact Pearlz support.

Code	Audible alert pattern	Meaning	What to do
E-08	Short tone every 2 seconds for 1 minute	Display and control board are not communicating	Unplug for 1 minute, then plug in again. If the code returns, contact Pearlz support.

To silence an audible alert, tap any button. The alert stays on the display until its cause is resolved.

14. Storage, moving and disposal

14.1 Storing the purifier

Before storing the purifier for more than 30 days, close the feed-water valve, dispense all water, unplug it and leave the filter compartment door open for a day so the compartment can dry. Store it upright in a dry place between 5 and 40 °C. When you start using it again, replace the PF-1 pre-filter and run the first-run flushing procedure in Section 6 of the AquaPearl Installation Guide.

14.2 Moving the purifier

Drain the tank and disconnect the tubes before moving the purifier. Keep it upright. Pearlz technicians can move and reinstall a purifier as part of a service visit; see the Service & Maintenance Agreement for the relocation charge.

14.3 Disposal

Do not put the purifier or used cartridges in household waste. Pearlz technicians collect used cartridges free of charge during service visits. At the end of the purifier's life, contact Pearlz support for collection and recycling.

15. Warranty and support

The AquaPearl 500 Pro is covered by a warranty of 24 months for the purifier body, pump and electronics. The warranty period starts on the date of your invoice. Filter cartridges are consumables and are not covered by the warranty. The full terms, including exclusions and how to make a claim, are in the Pearlz Warranty Policy.

When you contact support, have your invoice or order number and the serial number from the rating label ready. You can also find the serial number in the app under Settings, then Device info.

Channel	Details
Email	support@pearlzhome.example
Toll-free phone	0800-73275, Monday to Saturday, 9:00 to 18:00
WhatsApp	+92 300 0732759, every day, 9:00 to 21:00
App	Pearlz Home app, Help and Service